

Caleb James Murray

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Education

University of Tennessee, Knoxville – Bachelor of Science in Computer Engineering, 3.2 GPA.

Expected May 2027

Skills

PCB design: Kicad, Altium, Impedance control, high-speed signal integrity, and power management circuit design.

Embedded Systems: STM32, SPI/I2C/UART, Linux, Git, JTAG, Multithreading, Verilog, VHDL, C, C++, Python, Matlab.

Radar and RF: Pulse-Doppler and phased-array radar systems; RF PCB design, beamforming algorithms, network analyzers, oscilloscopes, power meters, spectrum analyzers, CST Studio Suite, Ansys HFSS, microwave circuits.

Experience

Radar Engineer, EWR Radar Systems – St. Louis, Missouri

April 2024 – Present

- Designed and tested a custom amplifier interface PCB that receives trigger signals from the signal processor and controls RF signal pass-through to the radar amplifier for transmit/standby operation; implemented gigabit Ethernet communication for improved signal integrity, 4-channel ADC for amp monitoring, and controlled impedance design; consolidated three boards into a single solution with 40% size reduction and significant cost savings.
- Designed and tested C-band and X-band active downconverter modules featuring IQ mixer architecture, integrated limiter/LNA front ends, and custom GaAs bias sequencing circuits; achieved ~1.67 dB system noise figure on Rogers 4350B substrate with +33 dBm CW input survivability and 40dB gain.
- Trained 20+ customers on radar operation and maintenance across domestic and international sites.

Student Research Assistant, Microwave Research Group – Knoxville, Tennessee

Jan 2026 – Present

- Designed a custom flight controller with an STM32H742 microcontroller and developed a complete software suite in C; implemented PID control algorithms using Simulink for autonomous stabilization and waypoint navigation.
- Developed a drone swarm coordination system enabling synchronized operation of up to 15 autonomous quadcopters with distributed communication protocols and collision avoidance algorithms.
- Designed phased array patch antennas and implemented beamforming algorithms for directional RF communication.

Projects

NFC-Enabled PCB Business Card

github.com/CalebJM04/Business-Card

- Designed a 2-layer PCB business card with integrated NFC technology and 1KB storage

Autonomous Rocket Flight Computer and Guidance System

github.com/CalebJM04/IEEE-2026-UAV

- Designed flight computer PCB with STM32 microcontroller integrating 9-DOF sensor suite (gyroscope, magnetometer, accelerometer, barometer) for autonomous guidance with aft-fin actuation and a custom control algorithm.
- Developed a custom RF telemetry link for real-time ground station communication.

Leadership and Extracurriculars

President, Tennessee Crew (Rowing)

August 2023-January 2026

- First sophomore president in club history; grew team from 12 to 32 members through recruiting initiatives and operational improvements; achieved 5 medals, including first place at Head of the South Regatta.

Vice-President and Founder, Onewheel Club

April 2024-Present

- Founded and scaled to 30+ active members within the first year; organized weekly rides and safety training sessions.

IEEE Robotics Team Member

August 2024-Present

- Designed custom flight controller PCB with STM32F405 and developed a complete firmware suite in C; integrated 915 MHz RF telemetry, RTK positioning, ESC control, and machine learning-enabled target recognition camera.

President, Chemical Engineering Car Club

August 2025-Present

- Grew club membership from 8 to 22 members; designed an analog optical differentiator circuit to monitor chemical reaction rate-of-change for an autonomous vehicle stopping mechanism, enabling precise distance control.

